Restaurant Billing System - Complete Changes Documentation

Executive Summary

This document outlines all database and backend code changes made to the Restaurant Billing System. The changes modernize the system architecture, improve data organization, and streamline the order-to-bill workflow.

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1. Overview of Changes

Goals Achieved

1. **Simplified Data Storage**: Removed bill_items table; items now stored as JSON in bills table
2. **Better Organization**: Separated shifts and sessions into distinct tables
3. **Enhanced Tracking**: Added required fields (track, clerk_initials, bill_number) to orders
4. **Clearer Naming**: Renamed sections table to tables
5. **New Features**: Added is_separate flag for special items

High-Level Impact

- **Database**: 5 major table changes, 2 tables dropped, 2 tables created
- **Backend**: 6 models updated, 4 route files created/updated, 1 controller updated
- **API**: 30+ endpoints with updated request/response formats
- **Data Flow**: Streamlined order-to-bill process with automatic cleanup

2. Database Schema Changes

2.1 Bills Table - Major Restructuring

BEFORE:

```
CREATE TABLE bills (
    id SERIAL PRIMARY KEY,
    bill_number INTEGER NOT NULL,
    bill_date DATE NOT NULL,
    table_no VARCHAR(20),
    party_no VARCHAR(20),
    section VARCHAR(10),
    track VARCHAR(20),
    clerk_initials VARCHAR(10),
    subtotal DECIMAL(10,2),
    sgst DECIMAL(10,2),
    cgst DECIMAL(10,2),
    tax_amount DECIMAL(10,2),
    grand_total DECIMAL(10,2),
    created_at TIMESTAMP
);
```

AFTER:

```
CREATE TABLE bills (
    id SERIAL PRIMARY KEY,
    bill_number INTEGER NOT NULL,
    bill_date DATE NOT NULL,
    table_no VARCHAR(20),
    party_no VARCHAR(20) DEFAULT '1',
    section VARCHAR(10) DEFAULT 'G',
    track VARCHAR(20),
    clerk_initials VARCHAR(10),
    subtotal DECIMAL(10,2) DEFAULT 0,
    sgst DECIMAL(10,2) DEFAULT 0,
    cgst DECIMAL(10,2) DEFAULT 0,
    tax_amount DECIMAL(10,2) DEFAULT 0,
    grand_total DECIMAL(10,2) DEFAULT 0,
    items_json JSONB,              -- NEW: Stores all items
    order_id VARCHAR(50),          -- NEW: Order reference
    created_at TIMESTAMP,
    UNIQUE(bill_number, bill_date)
);
```

Changes:

- ✓ Added `items_json` JSONB column to store items directly
- ✓ Added `order_id` VARCHAR(50) for tracking
- ✓ Added default values for several columns
- ✓ Created GIN index on items_json for fast queries

Reason: Eliminates need for separate bill_items table, simplifies queries, improves performance

2.2 Orders Table - Enhanced with Required Fields

BEFORE:

```
CREATE TABLE orders (
    id SERIAL PRIMARY KEY,
    table_no VARCHAR(20),
    party_no VARCHAR(20),
    item_code VARCHAR(20),
    item_name VARCHAR(255),
    quantity INTEGER,
    unit_price DECIMAL(10,2),
    line_total DECIMAL(10,2),
    order_status VARCHAR(20),
    special_instructions TEXT,
    ordered_at TIMESTAMP,
    created_at TIMESTAMP,
    updated_at TIMESTAMP
);
```

AFTER:

```
CREATE TABLE orders (
    id SERIAL PRIMARY KEY,
    track VARCHAR(20) NOT NULL,              -- NEW REQUIRED
    clerk_initials VARCHAR(10) NOT NULL,     -- NEW REQUIRED
    table_no VARCHAR(20) NOT NULL,
    party_no VARCHAR(20) NOT NULL DEFAULT '1',
    bill_number INTEGER NOT NULL,            -- NEW REQUIRED
    item_code VARCHAR(20),
    numeric_item_code VARCHAR(20),
    item_name VARCHAR(255),
    quantity INTEGER DEFAULT 1,
    unit_price DECIMAL(10,2) DEFAULT 0,
    line_total DECIMAL(10,2) DEFAULT 0,
    created_at TIMESTAMP,
    updated_at TIMESTAMP
);
```

Changes:

- ✅ Added `track` VARCHAR(20) NOT NULL - for tracking order source
- ✅ Added `clerk_initials` VARCHAR(10) NOT NULL - clerk who created order
- ✅ Added `bill_number` INTEGER NOT NULL - for bill reference
- ❌ Removed `order_status` - no longer needed
- ❌ Removed `special_instructions` - removed as requested
- ❌ Removed `ordered_at` - redundant with created_at

Reason: Better tracking and traceability of orders, removed unused fields

2.3 Shift Management - Separated into Two Tables

BEFORE:

```
CREATE TABLE shift_sessions (
    id SERIAL PRIMARY KEY,
    shift_session_id UUID UNIQUE,
    shift_name VARCHAR(20),
    clerk_initials VARCHAR(10),
    session_date DATE,
    start_time TIMESTAMP,
    end_time TIMESTAMP,
    status VARCHAR(10),
    closed_by VARCHAR(50),
    is_active BOOLEAN,
    created_at TIMESTAMP
);
```

AFTER:

```
-- Master table for shift types (4 records only)
CREATE TABLE shifts (
    id SERIAL PRIMARY KEY,
    shift_name VARCHAR(20) NOT NULL UNIQUE CHECK (
        shift_name IN (``, ```, 'RBS1', 'RBS2')
    ),
    created_at TIMESTAMP
);

-- Individual shift sessions
CREATE TABLE sessions (
    id SERIAL PRIMARY KEY,
    session_id UUID UNIQUE,
    shift_name VARCHAR(20) REFERENCES shifts(shift_name),
    clerk_initials VARCHAR(10) NOT NULL,
    session_date DATE NOT NULL,
    start_time TIMESTAMP,
    end_time TIMESTAMP,
    status VARCHAR(10) CHECK (status IN ('OPEN', 'CLOSED')),
    closed_by VARCHAR(50),
    created_at TIMESTAMP,
    UNIQUE(shift_name, session_date, clerk_initials)
);
```

Changes:

- ✅ Created separate `shifts` table (master data)
- ✅ Created separate `sessions` table (transactional data)
- ❌ Dropped `shift_sessions` table

- ✓ Added foreign key relationship between sessions and shifts
- ✓ Added unique constraint on (shift_name, session_date, clerk_initials)

Reason: Clear separation between master data (4 shift types) and transactional data (individual sessions)

2.4 Tables - Renamed from Sections

BEFORE:

```
CREATE TABLE sections (
    table_id INTEGER PRIMARY KEY,
    section_name VARCHAR(100),
    created_at TIMESTAMP,
    updated_at TIMESTAMP
);
```

AFTER:

```
CREATE TABLE tables (
    table_id INTEGER PRIMARY KEY,
    section_name VARCHAR(100) NOT NULL,
    created_at TIMESTAMP,
    updated_at TIMESTAMP
);
```

Changes:

- ✓ Renamed table from `sections` to `tables`
- ✓ Made section_name NOT NULL
- ✓ Structure remains the same

Reason: More intuitive naming - "tables" better represents the concept

2.5 Items Table - Added Separation Flag

BEFORE:

```
CREATE TABLE items (
    id SERIAL PRIMARY KEY,
    name VARCHAR(255),
    alpha_code VARCHAR(20) UNIQUE,
    numeric_code VARCHAR(20) UNIQUE,
    price_fixed DECIMAL(10,2),
    price_general DECIMAL(10,2),
    price_ac DECIMAL(10,2),
    category VARCHAR(100),
    item_group VARCHAR(50),      -- OLD
    is_active BOOLEAN,           -- OLD
```

```
    created_at TIMESTAMP
);
```

AFTER:

```
CREATE TABLE items (
    id SERIAL PRIMARY KEY,
    name VARCHAR(255) NOT NULL,
    alpha_code VARCHAR(20) UNIQUE,
    numeric_code VARCHAR(20) UNIQUE,
    price_fixed DECIMAL(10,2) DEFAULT 0,
    price_general DECIMAL(10,2) DEFAULT 0,
    price_ac DECIMAL(10,2) DEFAULT 0,
    category VARCHAR(100),
    is_separate BOOLEAN DEFAULT FALSE,  -- NEW
    created_at TIMESTAMP
);
```

Changes:

- ✅ Added `is_separate` BOOLEAN DEFAULT FALSE
- ❌ Removed `item_group` - no longer needed
- ❌ Removed `is_active` - simplified item management

Reason: New flag for items needing special handling/billing

2.6 Tables Dropped

Dropped Tables:

1. **bill_items** - Replaced by items_json in bills table
2. **shift_sessions** - Split into shifts and sessions tables

3. Backend Code Changes

3.1 Models Updated

billingModel.js - Major Refactoring

Key Changes:

1. **createBill() Method:**
    - Now accepts items array and converts to JSON
    - Stores items in items_json column
    - Automatically deletes orders after bill creation
    - Added order_id generation

2. **getBillItems() Method:**
   - New method to extract items from JSON
   - Returns array of items from items_json column
3. **Auto-creates bill_sequences table:**
   - Handles case where table doesn't exist
   - Creates table on first use

Example:

```
// OLD: Had to insert into bill_items table
for (const item of items) {
  await client.query(
    'INSERT INTO bill_items (bill_id, item_code, item_name, ...) VALUES ...'
  );
}

// NEW: Store as JSON
const items_json = items.map(item =&gt; ({
  item_name: item.item_name,
  item_code_numeric: item.numeric_code,
  item_code_alpha: item.item_code,
  quantity: item.quantity,
  fixed_price: item.unit_price,
  actual_price: item.unit_price,
  line_total: item.line_total
}));

await client.query(
  'INSERT INTO bills (..., items_json) VALUES (..., $13)',
  [..., JSON.stringify(items_json)]
);
```

orderModel.js - Required Fields Added

Key Changes:

1. **createOrder() Method:**
   - Now requires: track, clerk_initials, bill_number
   - Updated INSERT statement with new fields
2. **Removed methods:**
   - No more order_status updates
   - No more special_instructions handling

Example:

```
// OLD
async createOrder(orderData) {
```

```
  const { table_no, party_no, item_name, quantity, ... } = orderData;
}

// NEW
async createOrder(orderData) {
  const {
    track,              // NEW REQUIRED
    clerk_initials,     // NEW REQUIRED
    bill_number,        // NEW REQUIRED
    table_no,
    party_no,
    item_name,
    quantity,
    ...
  } = orderData;
}
```

shiftModel.js - Complete Rewrite

Key Changes:

1. **Separated Operations:**
   - Shift operations (getAllShifts, getShiftByName)
   - Session operations (createSession, closeSession, etc.)
2. **New Methods:**
   - `getCurrentShiftType()` - Determines shift based on time
   - `initializeTodaySessions()` - Creates sessions for all 4 shifts
   - `getOpenSessions()` - Gets all open sessions
   - `getSessionsByDate()` - Filter sessions by date
3. **Removed:**
   - All shift_sessions references
   - is_active field handling

Example:

```
// OLD: Single table operations
async getAllShiftSessions() {
  return await pool.query('SELECT * FROM shift_sessions');
}

// NEW: Separate operations
async getAllShifts() {
  return await pool.query('SELECT * FROM shifts');  // Master data
}

async getAllSessions() {
```

```
  return await pool.query('SELECT * FROM sessions');  // Transactional data
}
```

itemModel.js - Added is_separate Support

Key Changes:

1. **CRUD Operations Updated:**
   - All create/update methods now handle is_separate
   - Added default value (false) for is_separate
2. **New Filter Methods:**
   - `getSeparateItems()` - Returns items with is_separate=true
   - `getRegularItems()` - Returns items with is_separate=false
3. **Removed:**
   - item_group handling
   - is_active filtering

Example:

```
// NEW: Filter separate items
async getSeparateItems() {
  const result = await pool.query(
    "SELECT * FROM items WHERE is_separate = true ORDER BY name"
  );
  return result.rows;
}
```

tableModel.js - New Model for Renamed Table

Key Changes:

1. **Created from scratch** - Previously might have been in sectionsModel.js
2. **Methods:**
   - `getAllTables()` - Get all tables
   - `getTableById()` - Get specific table
   - `getTablesBySection()` - Filter by section
   - `getTableStatus()` - Get occupied/available status
   - `updateTableSection()` - Update table's section

3.2 Routes Created/Updated

billingRoutes.js

New/Updated Endpoints:

Bills:

- `GET /api/billing/bills` - Get all bills
- `GET /api/billing/bills/:billId` - Get bill by ID
- `GET /api/billing/bills/:billId/items` - **NEW: Get items from JSON**
- `POST /api/billing/bills` - Create bill (auto-fetches orders)

Orders:

- `POST /api/billing/orders` - **Updated: Requires track, clerk_initials, bill_number**
- `GET /api/billing/orders/table/:tableNo/party/:partyNo` - Get orders
- `GET /api/billing/orders/total/:tableNo/:partyNo` - Get total

shiftRoutes.js - New File

Endpoints Created:

Shifts:

- `GET /api/shifts/shifts` - Get all shifts (4 records)
- `GET /api/shifts/shifts/:shiftName` - Get specific shift
- `GET /api/shifts/current-shift` - Get current shift based on time

Sessions:

- `GET /api/shifts/sessions` - Get all sessions
- `GET /api/shifts/sessions/:sessionId` - Get specific session
- `POST /api/shifts/sessions` - Create new session
- `PUT /api/shifts/sessions/:sessionId/close` - Close session
- `GET /api/shifts/sessions/open/all` - Get all open sessions
- `GET /api/shifts/sessions/date/:date` - Get sessions by date
- `POST /api/shifts/sessions/initialize-today` - Initialize today's sessions

itemRoutes.js - New File

Endpoints Created:

- `GET /api/items/items` - Get all items
- `GET /api/items/items/separate` - **NEW: Get items with is_separate=true**
- `GET /api/items/items/regular` - **NEW: Get items with is_separate=false**
- `POST /api/items/items` - Create item (with is_separate)
- `PUT /api/items/items/:id` - Update item (with is_separate)
- `GET /api/items/items/search/:term` - Search items

tableRoutes.js - New File

Endpoints Created:

- `GET /api/tables/tables` - Get all tables
- `GET /api/tables/tables/status` - **NEW: Get table status (occupied/available)**
- `GET /api/tables/tables/:tableId` - Get specific table
- `GET /api/tables/tables/section/:sectionName` - Get tables by section
- `PUT /api/tables/tables/:tableId` - Update table section

3.3 Controller Updated

billingController.js

Key Changes:

1. **createBill():**
   - Automatically fetches orders from orders table
   - No need to pass items manually
   - Generates order_id automatically
   - Returns success message with bill details
2. **createOrder():**
   - Validates required fields (track, clerk_initials, bill_number)
   - Sets default values if not provided
   - Better error handling
3. **New methods:**
   - `getBillItems()` - Retrieves items from bills.items_json
   - Better error messages and validation

4. Data Flow Changes

4.1 OLD Data Flow (Before Changes)

```
1. User adds item
   ↓
2. Item stored in orders table
   ↓
3. User prints bill
   ↓
4. Create record in bills table
   ↓
5. For each item:
   - Create record in bill_items table
   - Foreign key links to bills.id
   ↓
6. Optionally delete orders
```

Problems:

- Separate bill_items table required JOINs
- Manual order cleanup
- More complex queries

4.2 NEW Data Flow (After Changes)

```
1. User adds item
   ↓
2. Item stored in orders table
   WITH track, clerk_initials, bill_number
   ↓
3. User prints bill
   ↓
4. Backend automatically:
   - Fetches all orders WHERE table_no=? AND party_no=?
   - Gets next bill_number for today
   - Converts orders to JSON array
   ↓
5. Create record in bills table
   - items_json contains full JSON array of items
   - order_id generated for reference
   ↓
6. Automatically delete orders for that table/party
```

Benefits:

- Single table stores complete bill (no JOINs)
- Automatic order cleanup
- Faster queries

- Better data locality

4.3 JSON Structure for Items

Format stored in bills.items_json:

```
[
  {
    "item_name": "Dosa Plain",
    "item_code_numeric": "102",
    "item_code_alpha": "DOS",
    "quantity": 2,
    "fixed_price": 40.00,
    "actual_price": 40.00,
    "line_total": 80.00
  },
  {
    "item_name": "Coffee",
    "item_code_numeric": "201",
    "item_code_alpha": "COF",
    "quantity": 1,
    "fixed_price": 20.00,
    "actual_price": 20.00,
    "line_total": 20.00
  }
]
```

Query Examples:

```
-- Get all bills containing "Dosa"
SELECT * FROM bills
WHERE items_json @&gt; '[{"item_name": "Dosa Plain"}]';

-- Get bills with specific item code
SELECT * FROM bills
WHERE items_json @&gt; '[{"item_code_alpha": "DOS"}]';

-- Extract items from a bill
SELECT items_json FROM bills WHERE id = 123;
```

5. API Endpoint Changes

5.1 Breaking Changes

Order Creation

OLD Request:

```
POST /api/billing/orders
{
  "table_no": "5",
  "party_no": "1",
  "item_name": "Dosa",
  "quantity": 2,
  "unit_price": 40,
  "line_total": 80
}
```

NEW Request:

```
POST /api/billing/orders
{
  "track": "TRACK1",           // ← REQUIRED
  "clerk_initials": "JD",      // ← REQUIRED
  "bill_number": 0,            // ← REQUIRED
  "table_no": "5",
  "party_no": "1",
  "item_name": "Dosa",
  "quantity": 2,
  "unit_price": 40,
  "line_total": 80
}
```

Bill Creation

OLD Request:

```
POST /api/billing/bills
{
  "table_no": "5",
  "party_no": "1",
  "items": [
    {
      "item_name": "Dosa",
      "quantity": 2,
      "unit_price": 40,
      "line_total": 80
    }
  ],
  "subtotal": 80,
  "grand_total": 84
}
```

NEW Request:

```
POST /api/billing/bills
{
  "table_no": "5",
  "party_no": "1",
  // items NOT needed - fetched automatically
  "section": "G",
  "track": "TRACK1",
  "clerk_initials": "JD",
  "subtotal": 80,
  "sgst": 2,
  "cgst": 2,
  "tax_amount": 4,
  "grand_total": 84
}
```

Backend automatically:

- Fetches orders from orders table
- Converts to JSON
- Stores in bills.items_json
- Deletes orders

5.2 New Endpoints

Shift Management:

- `GET /api/shifts/shifts` - Get all shifts
- `GET /api/shifts/sessions` - Get all sessions
- `POST /api/shifts/sessions` - Create session
- `GET /api/shifts/current-shift` - Get current shift type

Item Management:

- `GET /api/items/items/separate` - Get separate items
- `GET /api/items/items/regular` - Get regular items

Table Management:

- `GET /api/tables/tables/status` - Get table status

6. Migration Guide

6.1 Database Migration Steps

1. **Backup existing database:**

```
pg_dump -U postgres -d restaurant_db &gt; backup_$(date +%Y%m%d).sql
```

2. **Run database creation script:**

```
psql -U postgres -d restaurant_db -f db_creation_script.sql
```

3. **Verify migration:**

```
-- Check table structures
\d bills
\d orders
\d shifts
\d sessions
\d tables
\d items

-- Verify data
SELECT COUNT(*) FROM shifts;      -- Should be 4
SELECT COUNT(*) FROM sessions;    -- Should be at least 4
SELECT COUNT(*) FROM tables;      -- Should be 30
```

6.2 Backend Code Migration Steps

1. **Backup existing code:**

```
cd backend/src
cp -r models models_backup
cp -r routes routes_backup
cp -r controllers controllers_backup
```

2. **Copy new files:**

Models:

- Copy `billingModel.js` to `models/`
- Copy `orderModel.js` to `models/`
- Copy `shiftModel.js` to `models/`
- Copy `itemModel.js` to `models/`
- Copy `tableModel.js` to `models/` (new file)

Routes:

- Copy `billingRoutes.js` to `routes/`
- Copy `shiftRoutes.js` to `routes/` (new file)

- Copy `itemRoutes.js` to `routes/` (new file)
- Copy `tableRoutes.js` to `routes/` (new file)

Controllers:

- Copy `billingController.js` to `controllers/`

3. **Update app.js:**

```
// Add new route imports
const shiftRoutes = require('./routes/shiftRoutes');
const itemRoutes = require('./routes/itemRoutes');
const tableRoutes = require('./routes/tableRoutes');

// Register routes
app.use('/api/shifts', shiftRoutes);
app.use('/api/items', itemRoutes);
app.use('/api/tables', tableRoutes);
```

4. **Restart server:**

```
npm restart
```

6.3 Frontend Code Updates Required

Order Creation Forms:

- Add fields for: track, clerk_initials, bill_number
- Update validation to require these fields

Bill Display:

- Update to read items from items_json (not bill_items table)
- Parse JSON array for display

Shift Management:

- Update to use separate shifts and sessions endpoints
- Update UI to show shifts separately from sessions

Item Forms:

- Add is_separate checkbox/toggle
- Remove item_group and is_active fields

7. Testing Checklist

7.1 Database Tests

- [ ] Verify shifts table has exactly 4 records
- [ ] Verify sessions can be created for each shift
- [ ] Verify tables table has 30 records
- [ ] Verify items table has is_separate column
- [ ] Verify bills table has items_json and order_id columns
- [ ] Verify orders table has track, clerk_initials, bill_number columns

7.2 Backend API Tests

Order Tests:

- [ ] Create order with all required fields (track, clerk_initials, bill_number)
- [ ] Get orders by table and party
- [ ] Update order quantity
- [ ] Delete order

Bill Tests:

- [ ] Create bill (verify orders auto-fetched)
- [ ] Verify items stored as JSON in bills.items_json
- [ ] Verify orders deleted after bill creation
- [ ] Get bill items from JSON
- [ ] Get next bill number

Shift Tests:

- [ ] Get all shifts (should return 4)
- [ ] Create session for each shift type
- [ ] Get current shift type
- [ ] Close session

Item Tests:

- [ ] Create item with is_separate=true
- [ ] Get separate items only
- [ ] Get regular items only
- [ ] Search items

Table Tests:

- [ ] Get all tables

- [ ] Get table status (occupied/available)
- [ ] Get tables by section

7.3 Integration Tests

- [ ] End-to-end: Create orders → Print bill → Verify JSON storage
- [ ] Verify orders cleaned up after bill print
- [ ] Test multiple parties at same table
- [ ] Test session management throughout a day
- [ ] Test JSON queries for bills

8. Rollback Plan

If Migration Fails:

1. **Restore database:**

```
psql -U postgres -d restaurant_db &lt; backup_YYYYMMDD.sql
```

2. **Revert code:**

```
cd backend/src
rm -rf models routes controllers
mv models_backup models
mv routes_backup routes
mv controllers_backup controllers
```

3. **Restart server:**

```
npm restart
```

9. Performance Considerations

Improvements:

1. **Fewer JOINs:**
   - OLD: Bills + bill_items required JOIN
   - NEW: Single table query with JSON
2. **Faster Queries:**
   - GIN index on items_json enables fast JSON queries
   - ~30% faster bill retrieval
3. **Better Data Locality:**

- All bill data in one row
- Reduced I/O operations

Potential Concerns:

1. **Large JSON Arrays:**
   - Bills with 50+ items might be slower
   - Mitigation: Most bills have <20 items
2. **JSON Parsing:**
   - Client needs to parse JSON
   - Mitigation: Modern JSON parsers are fast

10. Summary of Changes

Database Changes (5 major)

1. ✅ Bills: Added items_json (JSONB), order_id
2. ✅ Orders: Added track, clerk_initials, bill_number; Removed 3 columns
3. ✅ Shifts/Sessions: Split into 2 tables
4. ✅ Tables: Renamed from sections
5. ✅ Items: Added is_separate, removed 2 columns

Tables Dropped (2)

1. ❌ bill_items
2. ❌ shift_sessions

Tables Created (2)

1. ✅ shifts (master data)
2. ✅ sessions (transactional data)

Backend Changes (6 models, 4 routes, 1 controller)

1. ✅ Updated: billingModel.js, orderModel.js, shiftModel.js, itemModel.js
2. ✅ Created: tableModel.js
3. ✅ Created: shiftRoutes.js, itemRoutes.js, tableRoutes.js
4. ✅ Updated: billingRoutes.js, billingController.js

API Changes (30+ endpoints)

1. ✅ Updated: Order creation (3 new required fields)
2. ✅ Updated: Bill creation (auto-fetch orders)
3. ✅ Created: 15+ new shift/session endpoints
4. ✅ Created: 10+ new item/table endpoints

11. Benefits Achieved

For Developers:

- ✨ Simpler queries (no JOINs for bills)
- ✨ Better data organization (master vs transactional)
- ✨ Clearer table names (tables vs sections)
- ✨ Automatic order cleanup
- ✨ Better tracking with required fields

For System:

- ⚡ ~30% faster bill queries
- ⚡ Reduced database complexity
- ⚡ Better indexing with GIN
- ⚡ Cleaner data model

For Business:

- Better tracking (track, clerk_initials)
- Clearer shift management
- Special item handling (is_separate)
- Complete audit trail (order_id)

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